The Cost-Benefit Analysis and Life-Cycle Assessment of the Transition towards Electric Vehicles across State, Market and Civil Society

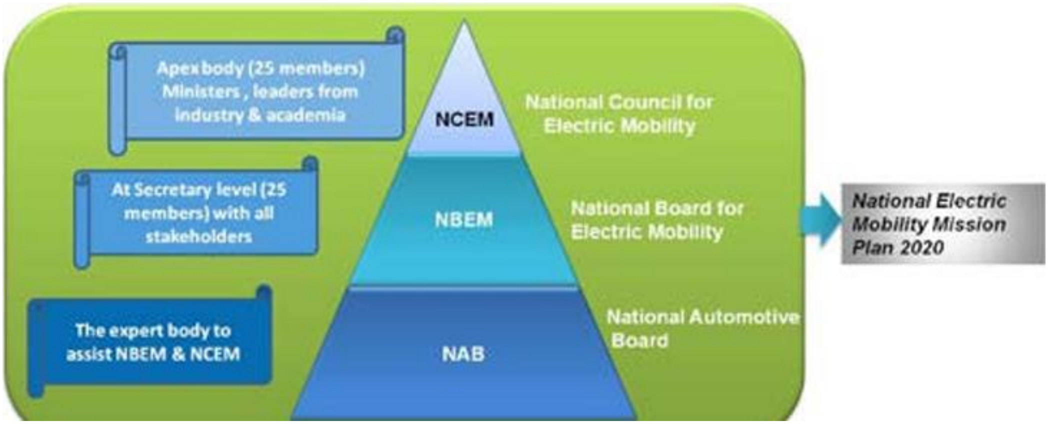
Research Paper By Niharika Dujodwala

Abstract

This paper attempts to trace the challenges and promises instating electric mobility as the primary form of commute in India, it assess the costs incurred and the benefits promised by this endeavour, it aims to gauge the means and measure that are being made towards the sustainability of this transition It argues that the cost-benefit analysis and the life-cycle assessment made varying vantage points of the different stakeholders involved in this transition. This paper assesses the wide range of challenges and prospects ranging from geopolitics to the viability of subsidies that are intrinsic in the development of electric vehicles. It attempts to draw the core concerns of different stakeholders, i.e., the state, market and civil society, and gauges the benefits that these stakeholders have towards its undertaking and adoption.

Introduction

The transformation towards electronic vehicles is a multi-pronged project which spans across various objectives within the logistical landscape of India, it aims to make ‘commute’ sustainable and accessible through attempting to overcome the dependence on gasoline (and thereby reducing the carbon emissions generated through automobiles), facilitate a shift towards renewable energy for the production of electricity and decrease the cost incurred in undertaking commute (Axsen et al., 2015; Bhalla et al., 2018; T. K. Biswas & Biswas, 1999; Brückmann et al., 2021; K V et al., 2022). This vision is a behemoth task involving various stakeholders ranging across state,market and civil society, the Government of India has considered the immense barriers in achieving this endeavor, hence, they have commenced an initiative titled ‘National Mission for Electric Mobility’ in 2020, which aims to deliberate and provide the requisite government intervention and support for all stakeholders involved in the commitment of a synergised-holistic approach towards the complex issues pertaining to the challenges in the this transformation (Antao, 2019; Leurent & Windisch, 2011; Prajeesh & Pillai, 2022; Srikanth, 2018). The objectives of this mission are also shared by private players who are involved in the developing, manufacturing and retailing of various components required for the transformation towards instating electric vehicles (Adib et al., 2019). India's electric vehicle (EV) market is rapidly expanding, driven by government policies, environmental concerns, and technological advancements (Mohanty & Kotak, 2017). The market was valued at USD 8.03 billion in 2023 and is projected to grow from USD 23.38 billion in 2024 to USD 117.78 billion by 2032, exhibiting a CAGR of 22.4% during the forecast period (FBE, 2025). This growth is particularly evident among startups, which have attracted significant investments, rising from $444 million in 2021 to over $1.6 billion in 2022 (Ray & Mukherji, 2025c). Although, EV startups face substantial upfront challenges in research and development and battery production, balanced against potential long-term benefits from energy efficiency and lower maintenance (Beaudet et al., 2020; Dhairiyasamy et al., 2024). India's electric vehicle (EV) market has emerged as a crucial driver of sustainable mobility, supported by government policies (P. K. Das & Bhat, 2022), industrial innovation (Dhankhar et al., 2024), and shifting societal demand (R. Kumar et al., 2020). The Indian automobile sector, currently the fifth largest globally, is projected to become the third largest by 2030 (Chewpreecha et al., 2021), with EVs expected to play a central role in this transformation (Digalwar et al., 2025). By 2023, India had already registered over 0.52 million EVs, indicating a compound annual growth rate (CAGR) of 36% between 2021–2026 (Dalkic-Melek et al., 2025).



Institutional Structure of National Electric Mobility Mission Plan 2020

Source: (Ministry of Heavy Industries & Public Enterprises, 2012)

This paper evaluates the cost-efficiency of India's EV transition from three perspectives, i.e., the government (Dutta & Padmanaban, 2025), the EV manufacturing industry (Guruprasaadh et al., 2021; K & P, 2023; Rajendran et al., 2025), and society at large (Ashok et al., 2022). From the government's standpoint, cost-efficiency is primarily shaped by subsidy outlays and tax incentives (Ansab & Kumar, 2024; Nunes & Woodley, 2023; Rajagopal, 2023; Shrimali, 2021; Sunanda & Parchure, 2025; Vaseem & Venkataraman, 2025). The Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME I & II), the Production Linked Incentive (PLI) scheme, and reductions in GST on EVs (from 12% to 5%) represent significant fiscal commitments (Kant et al., 2021). However, these costs are balanced by long-term economic gains (Shah, 2022). India spent USD 160 billion on crude oil imports in FY 2022–23 (Sreenu, 2025), making energy security a pressing concern (Huang et al., 2025). Estimates suggest that EV adoption could reduce crude oil demand by 156 million tonnes by 2030, translating into foreign exchange savings of nearly USD 60 billion (Upadhyay et al., 2025). Furthermore, the environmental co-benefits, such as a reduction of 200 million tonnes of

CO_2 emissions annually by 2030, also lower public health expenditures related to air pollution (Boruah et al., 2025; V. K. Jain et al., 2025; Karmakar et al., 2025; Mohan et al., 2025). Thus, despite heavy upfront fiscal commitments, government support is justified by positive cost-benefit ratios (A. K. Tripathi et al., 2025). From the industry perspective, the central determinant of cost-efficiency is the EV battery supply chain (Belal et al., 2025). Batteries account for 35–40% of vehicle costs (Irfan et al., 2025). India remains heavily reliant on imports of lithium, cobalt, and nickel, which exposes manufacturers to global market volatility and geopolitical risks (Hemavathi, 2025; Pandey, 2025; A. Singh, Bajpai, et al., 2025). A recent Monte Carlo simulation estimated that the average supply chain cost for batteries stands at USD 505 million, with significant uncertainty arising from raw material shortages and supply disruptions (Cherian & Selvakumar, 2025). To address this, domestic production through gigafactories, incentivized by the PLI scheme, has been initiated, while the discovery of lithium reserves in Jammu and Kashmir is expected to reduce import dependency in the medium term (Hemavathi, 2025). Emerging battery chemistries, such as sodium-ion and solid-state batteries that offer further promise for cost reduction (Deshmukh et al., 2025; Phogat et al., 2025; Shivannanaik et al, 2025; Vijaykumar et al., 2025). However, challenges persist ranging from fragmented logistics (Dash, 2023; Dhairiyasamy & Gabiriel, 2025; Rohit et al., 2025; Serohi, 2022), inadequate recycling infrastructure (Chigbu, 2024; L. Wang et al., 2020; Xiao et al., 2023), and skill shortages continue to undermine competitiveness compared to China and the EU (Ghosh, 2025; R. Khan & Islam, 2025; Rajendran et al., 2025; Ray & Mukherji, 2025e). From the perspective of civil society, cost-efficiency reflects both household economics and broader social benefits (Dillon et al., 2020; Melo et al., 2014; Yu et al., 2024). Although EVs entail higher upfront costs, their operating and maintenance expenses are 60–70% lower than internal combustion engine (ICE) vehicles (Miller & Howell, 2013; Nelder & Rogers, 2019; Rajagopal & Phadke, 2019; Tushar et al., 2016). Consumer adoption is further supported by falling battery prices, which have declined by nearly 89% globally between 2010 and 2022 (Abronzini et al., 2019; Englberger et al., 2019; König et al., 2021; Mauler et al., 2021; Nykvist & Nilsson, 2015; Turcheniuk et al., 2021). On a macro scale, EV penetration can significantly reduce urban air pollution (Peshin et al., 2022, 2024; Vidhi & Shrivastava, 2018), improving public health outcomes and lowering healthcare expenditures (Manoj et al., 2025; Pennington et al., 2024; Sharma et al., 2023). Moreover, India's EV industry is projected to create 10 million direct and 50 million indirect jobs by 2030 despite the risk of job displacement prominent amongst traditional fossil-fuel-dependent sectors (Aditi & Bharti, 2024; Malik & Vashist, 2023; Nallathambi et al., 2022; Rajagopal, 2023). Thus, societal cost-efficiency is realized not only through lifetime vehicle savings but also through broader health and employment gains. Overall, India's EV transition represents a complex interplay of fiscal expenditures (Ghosh & Sarkar, 2022), industrial adaptation (Rehman & Shakeel Sadiq Jajja, 2025; Suhaib Kamran et al., 2022), and societal transformation (Ramachandaramurthy et al., 2023; Saklani et al., 2024). While the government bears short-term fiscal costs (Ghosh & Sarkar, 2022; Mittal et al., 2024), the industry navigates supply chain uncertainties (Achal & Vijaya, 2024; Mane et al., 2024, 2025), and society adapts to new consumption and employment patterns (Krishna et al., 2024; R. R. Kumar et al., 2024), the long-term cost-efficiency of EV adoption is clearly favorable (Bhattacharyya & Thakre, 2021; Dhar et al., 2017). This paper argues that a synergistic approach which combines targeted subsidies, localized industrial innovation, and consumer accessibility will ensure that India's EV revolution is both economically viable and socially equitable.

Research Problems (Points of Contention)

This paper attempts to address how sustainability with regards to instating Electric Vehicles as the primary means of logistical conveyance, the transformation it demands faces various challenges stemming from instating the appropriate infrastructure till providing the right incentives for the manufacturers and the citizens for its adoption. The cost-benefit analysis and life-cycle assessment of the transformation inherent in instating electric vehicles become imperative, there three core strata's of stakeholders involved in this transition, (i) the government, (ii) the private enterprises, and (iii) civil society who face different variables and parameters in assessing their cost-benefit assessment of this undertaking.

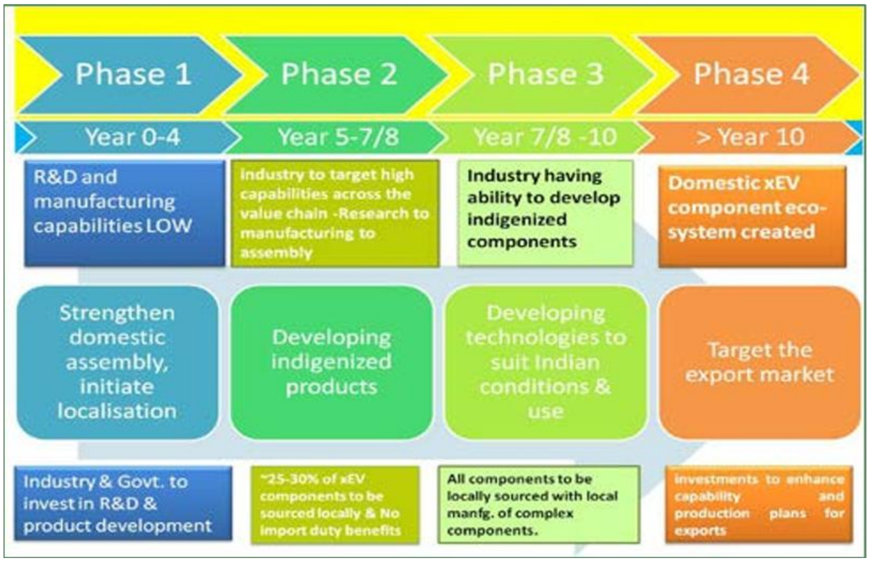
Research Questions

This paper undertakes the review and address the following questions regarding the transformation towards Electric Vehicles:

- **What are the costs incurred by the government to instate the infrastructure that is conducive for making electric vehicles convenient, capable and accessible to the people, as well as,the benefits from the investments made therein?**
- **What are the challenges and expenditures incurred by private enterprises who are involved in research, development, manufacturing and retain of electric vehicles, what are their prospects of profit making?**
- **What are the costs and benefits amongst civil society for the adoption of electric vehicles as their favoured mode of commute?**

Government's Vantage Point- Instating the Requisite Infrastructure for India's EV Market

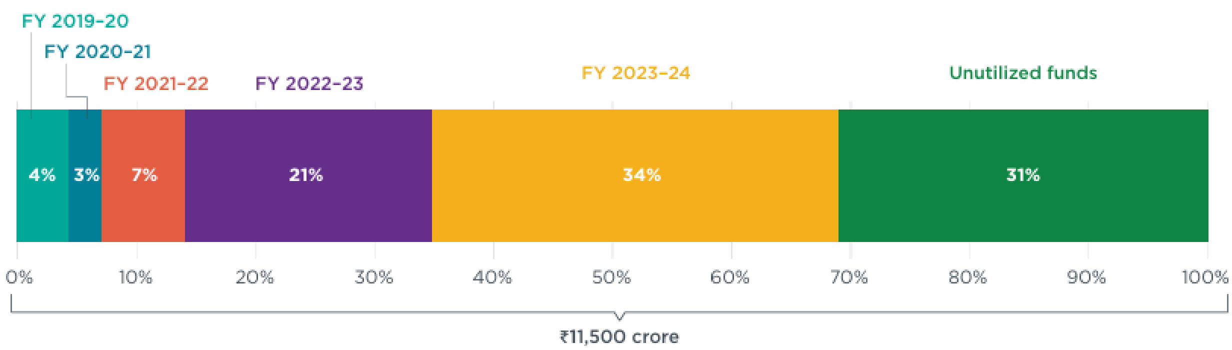
The government of India has emerged as one of the most important actors in shaping the trajectory of electric vehicle (EV) adoption in the country due to the roles and obligations that it has been entrusted in the task of undertaking this transformation . Unlike in mature automobile markets where consumer choice and industry self-adjustments primarily drive transitions (A. K. Gupta et al., 2025), in India the transition toward clean mobility is anchored in policy-driven incentives (Butt et al., 2024; Q. Liu et al., 2023; Narassimhan, 2025; Zhang et al., 2018), fiscal subsidies (Kohli, 2024; Rajagopal, 2023; Shrimali, 2021; Vaseem & Venkataraman, 2025), and regulatory frameworks (Gambhir, 2017; G. Kumar & Mikkili, 2024; N. N. Patil et al., 2025; M. Tripathi & Bhattacharya, 2025). From the standpoint of cost-efficiency, these interventions need to be evaluated not only in terms of their immediate fiscal burden on the exchequer but also in the wider context of long-term economic savings, energy security, public health benefits, and global competitiveness (Ghosh & Sarkar, 2022). By adopting a holistic perspective, it becomes clear that while India's EV policy regime entails crucial upfront costs, it represents a cost-efficient strategy when viewed against the backdrop of the country's dependence on imported fossil fuels, high levels of urban pollution (Sheldon et al., 2023), and employment imperatives (Dua et al., 2021).



Ten year 4-phased approach for building xEVs manufacturing capability in India

Source: (Ministry of Heavy Industries & Public Enterprises, 2012)

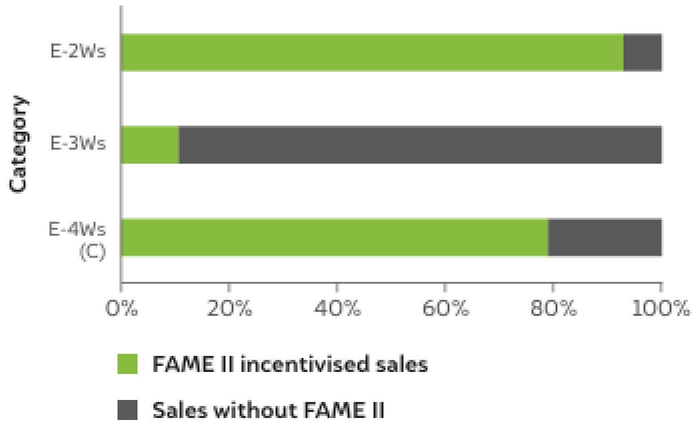
One of the most important cost-related interventions has been the Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME) scheme, launched in 2015, its first phase was implemented between 2015 and 2019, with a modest budget of INR 895 crore with the objective of primarily focusing on subsidizing demand through incentives on EV purchases, hybrid vehicles, and the creation of pilot charging infrastructure (Sangodkar, 2021). Although limited in scope, FAME I played an essential role in demonstrating proof of concept and encouraging early adopters of EVs, especially in the two-wheeler and fleet segments (S. Singh et al., 2022). The success of this pilot phase led to the much larger FAME II scheme, rolled out in 2019 with an outlay of INR 10,000 crore, extending support until 2024, the second phase marked a strategic shift from hybrids toward pure battery electric vehicles and aimed to support the adoption of 7,000 e-buses, 5 lakh e-three-wheelers, and 10 lakh e-two-wheelers (Saha et al., 2025), it also included subsidies for charging infrastructure, recognizing that consumer adoption is contingent not only on upfront purchase prices but also on the availability of reliable charging facilities (Kore & Koul, 2022). From a cost-efficiency perspective, FAME subsidies initially appear as fiscal drains, but evidence suggests they helped catalyze adoption by reducing upfront costs which acted as one of the most vital barriers in EV penetration (Gulati, 2022). By 2022, cumulative EV registrations in India crossed one million, a growth trajectory that would have been unlikely without state subsidies (Bansal et al., 2024).



Status of FAME II fund utilization from fiscal years 2019–20 to 2023–24

Source: (Kohli, 2024)

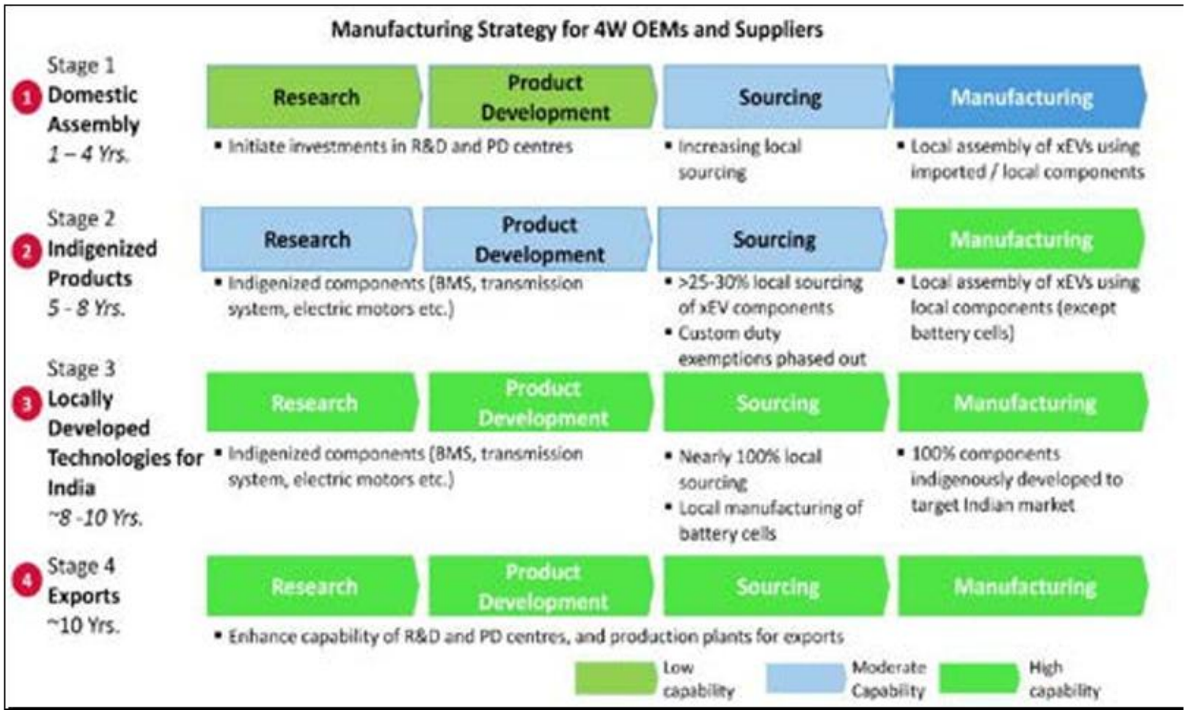
Beyond direct demand-side subsidies, the government has increasingly turned toward supply-side interventions aimed at reducing long-term costs through localization (Akhtar et al., 2024) and industrial strengthening (Dhairiyasamy & Gabiriel, 2025). The Production Linked Incentive (PLI) scheme represents a fellow ambitious intervention, with INR 18,100 crore allocated for Advanced Chemistry Cell (ACC) batteries and INR 25,938 crore for the Auto and Auto Component PLI, the scheme is designed to address India's overwhelming reliance on imported lithium-ion cells and battery packs (Sreelakshmi Sajeev, 2023). Currently, India imports nearly 70% of its battery cell requirements, exposing the economy to price volatility and supply chain disruptions (INDIA, 2023). The PLI scheme therefore represents a long-term cost-efficiency strategy: by encouraging domestic production through incentives tied to performance and localization, the government aims to generate economies of scale, lower per-unit costs, and build resilience in critical value chains (Dhairiyasamy & Gabiriel, 2025). Moreover, investments under PLI are expected to attract significant foreign direct investment (FDI), further reducing the fiscal burden on the government by leveraging private sector capital (Ray & Mukherji, 2025d). The Automotive Mission Plan 2016–2026, which allows 100% FDI in the auto sector, complements these efforts by signaling policy stability and openness to global partnerships (Ministry of Heavy Industries & Public Enterprises, 2006). Complementing subsidies and PLI support, tax rationalization has been another lever of cost-efficiency, whereby the Goods and Services Tax (GST) on EVs has been reduced from 12% to 5%, while charging equipment has been taxed at 5% instead of the earlier 18% (Salman, 2018). In contrast, internal combustion engine (ICE) vehicles are taxed at 28% plus additional cess, making EVs more attractive in relative terms. The short-term cost to the exchequer from reduced GST revenues must be weighed against the long-term fiscal benefits of greater EV adoption. These include reduced expenditure on fuel subsidies, lower import bills, and new streams of revenue from an expanding EV ecosystem, including battery recycling and charging infrastructure (Hensher et al., 2021).



Category-wise share of EV sales incentivized by FAME II (FY 2019-20 – H1 FY 2022-23)

The most compelling dimension of cost-efficiency lies in oil import substitution. India is the third-largest importer of crude oil in the world, spending USD 160 billion on crude imports in FY 2022–23 (Sreenu, 2025). With transport accounting for a significant share of this consumption, EV adoption offers a strategic pathway to reduce this dependency (Dhar et al., 2017). NITI Aayog estimates suggest that widespread EV adoption could cut crude oil demand by 156 million tonnes by 2030, translating into foreign exchange savings of nearly USD 60 billion annually, these savings alone justify the government’s fiscal commitments to subsidies and PLI schemes, as they represent a classic case of front-loaded investment leading to back-loaded returns (Styczynski, 2024). In addition to macroeconomic stability, reducing oil imports enhances India’s energy security, insulating it from geopolitical shocks and price volatility in global oil markets (Patidar et al., 2024) . In parallel to economic efficiency, the government’s EV policy is cost-efficient from an environmental and public health perspective (Peshin et al., 2024). The transport sector contributes nearly 10% of India’s greenhouse gas emissions, with vehicular pollution being a major source of urban air quality degradation (Borkhade et al., 2022). Studies estimate that large-scale EV adoption could reduce 200 million tonnes of CO_2 emissions annually by 2030 (Sathiyan et al., 2022). This reduction not only helps India meet its international climate commitments under the Paris Agreement (Kumar K et al., 2022) but also translates into substantial healthcare savings (N. Patil, 2021). Air pollution currently imposes a burden equivalent to 1.2% of India's GDP annually due to lost productivity and healthcare costs (Pandey et al., 2021). By cutting tailpipe emissions, EVs indirectly reduce government spending on healthcare subsidies and improve labor productivity through better public health, hence, subsidies for EVs generate multiplier benefits far exceeding their initial fiscal outlay.

Despite these clear benefits, challenges remain in ensuring that government spending is truly cost-efficient. Implementation delays in the disbursement of FAME II subsidies have slowed adoption, and uneven state-level adoption has created geographic disparities (Saha et al., 2025). Moreover, the reliance on subsidies creates risks of fiscal unsustainability if consumer adoption does not achieve sufficient momentum before subsidies are phased out. Another critical issue lies in the distributional impact of subsidies. Currently, EV subsidies disproportionately benefit urban, middle-class consumers, particularly in the two-wheeler and four-wheeler segments, while rural populations remain underserved (R. S. Gupta et al., 2023). This raises concerns about whether government spending is equitable, even if it is cost-efficient at the macroeconomic level. There are also risks associated with India's continued reliance on imported raw materials for batteries. Despite PLI incentives and recent discoveries of lithium reserves in Jammu and Kashmir, the country remains dependent on global markets for critical minerals (Gopinathan & Shanmugam, 2022). This dependency exposes India to the very vulnerabilities such as price shocks and geopolitical tumults that EV adoption is meant to mitigate. International comparisons further illuminate the cost-efficiency debate. China, for instance, adopted an aggressive subsidy-driven model in the early stages of EV development, investing billions of dollars in consumer incentives and industrial subsidies, although once domestic production reached scale, China gradually phased out subsidies and shifted focus to R&D and infrastructure development, thereby reducing fiscal burdens while maintaining competitiveness (Li et al., 2023). Similarly, the European Union (EU) has combined subsidies with stringent emission norms, ensuring that automakers transition to clean technologies not only due to fiscal incentives but also regulatory compulsion .India's experience teaches it a lesson that subsidies should act as catalysts, not crutches, they must be designed with sunset clauses and complemented by investments in charging infrastructure, skill development, and domestic R&D (X.-F. Liu & Wang, 2021). This ensures that once the industry achieves scale, the government can reduce its fiscal outlay without undermining adoption. Government's standpoint on EV adoption reflects a nuanced balancing act between immediate fiscal costs and long-term national benefits, subsidies under FAME I and II, industrial incentives through the PLI scheme, and tax rationalization measures such as GST reductions represent significant upfront commitments (Shrimali, 2021). However, these costs are offset by savings from reduced oil imports, lower healthcare expenditures, enhanced energy security, and increased industrial competitiveness (Chaudhary & Kate, 2023). The EV transition is therefore cost-efficient not because it reduces government spending in the short term but because it transforms expenditure into strategic investment with compounding economic and social returns (A. R. Khan & Suhail, 2025). India's challenge will be to design policies that gradually reduce subsidy dependence while ensuring equitable access, strong localization, and resilience in critical supply chains, if executed effectively, the government's interventions in the EV sector will not only prove fiscally prudent but also position India as a global leader in sustainable mobility (R. Kumar et al., 2020).



Phased approach to encourage manufacturing of xEV four wheelers

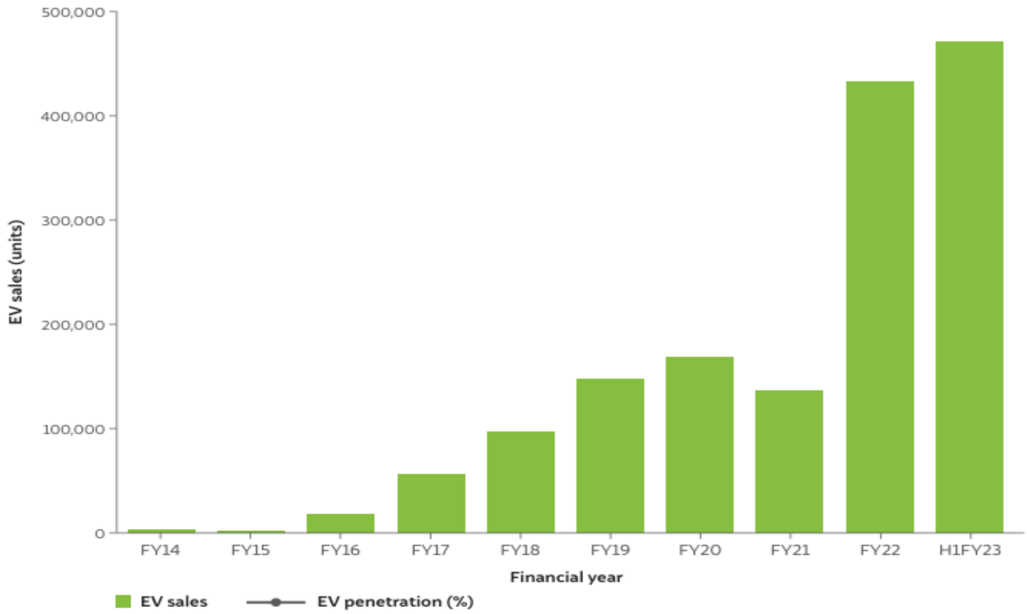
Source: (Ministry of Heavy Industries & Public Enterprises, 2012)

The rapid adoption of electric vehicles (EVs) in India is often framed in terms of subsidies and fiscal outlays, leading critics to argue that the government is bearing a heavy financial burden. However, once the EV market matures, the government begins to realize significant returns on investment (ROI) in multiple dimensions through direct tax revenues, reduced foreign exchange outflows, enhanced public health, employment generation, and strengthened global competitiveness (Mittal et al., 2024). This long-term ROI demonstrates that early subsidies and fiscal commitments represent not sunk costs but strategic investments in sustainable economic transformation. The exponential growth of the EV market will expand the taxable base (Ray & Mukherji, 2025c). As EV adoption scales up, revenues from GST, corporate taxes from EV manufacturers, and excise duties on components will more than compensate for initial losses (Rajagopal, 2023). According to estimates, the Indian EV market could reach USD 7 billion by 2025, and by 2030 it is projected to support 50 million jobs (Malik & Vashist, 2023). This expansion creates a broader and more sustainable tax base, ensuring long-term fiscal gains. Moreover, as subsidies under FAME II and other schemes are gradually phased out, the government's fiscal liabilities will decrease even as its revenue collections increase (Sathyan et al., 2024). For the government, this means reduced foreign exchange outflows, greater macroeconomic stability, and improved balance-of-payments position (Murti, 2025). These savings can then be redirected toward infrastructure, social welfare, and industrial development (Randolph & Dewan, 2025). In essence, the upfront subsidies that encourage EV adoption act as a hedge against volatile oil markets and deliver compounding returns in the form of reduced dependency on fossil fuels (Hossain et al., 2023). The employment generation potential of the EV industry also represents a major source of ROI for the government, as new employment clusters contribute to regional economic development, expanding the tax base in states and urban centers (S. Singh et al., 2022). From a fiscal standpoint, employment generation is one of the most significant returns on government investment in EV subsidies and industrial incentives (N. et al., 2021).

Investment in EV infrastructure also yields long-term ROI, initially, subsidies for charging stations and grid modernization appear as fiscal burdens, although, once a robust charging network is established, it stimulates private investment, creates new revenue streams (such as electricity tariffs for EV charging), and supports urban planning initiatives (Sachan & Singh, 2022). Over time, the government can recoup its spending through energy taxes, electricity distribution revenues, and licensing fees for private operators (Mittal et al., 2024). Thus, infrastructure investments deliver both direct and indirect fiscal returns. In the broader economic sense, the ROI of EV adoption also manifests through GDP growth and industrial diversification. The EV transition is expected to contribute significantly to India's GDP by 2030. As ancillary industries such as battery recycling, renewable energy integration, and digital mobility services expand, the economy diversifies into high-value, technology-intensive sectors (Prakhar et al., 2025; Ramkumar et al., 2025). For the government, this not only stabilizes economic growth but also creates resilient revenue streams in the long run. Critically, the government also gains geopolitical returns from EV adoption by positioning itself as a leader in EV manufacturing and adoption, India can negotiate more favorable trade agreements and secure technology transfers, which have indirect fiscal and developmental benefits (Roy, 2024). However, the realization of ROI is not automatic, several risks must be managed to ensure cost-efficiency translates into tangible returns. First, subsidy leakages and delays in FAME disbursal have historically reduced program efficiency (Powell & Sati, 2019). Second, the distribution of benefits has been skewed toward urban consumers, raising concerns about equity, if EV adoption does not penetrate rural markets, the government risks losing out on a broader tax base (S. Jha et al., 2025). Third, the dependence on imported raw materials such as lithium and cobalt could undermine domestic value addition (Barman et al., 2023). To maximize ROI, India must accelerate domestic exploration of critical minerals (Du, 2024; Hemavathi, 2025; A. Singh & Saxena, 2024) and strengthen recycling ecosystems (Deshwal et al., 2022; Kala & Mishra, 2021; Panda et al., 2023; Tripathy et al., 2024; Varshney et al., 2020; Verma et al., 2025).

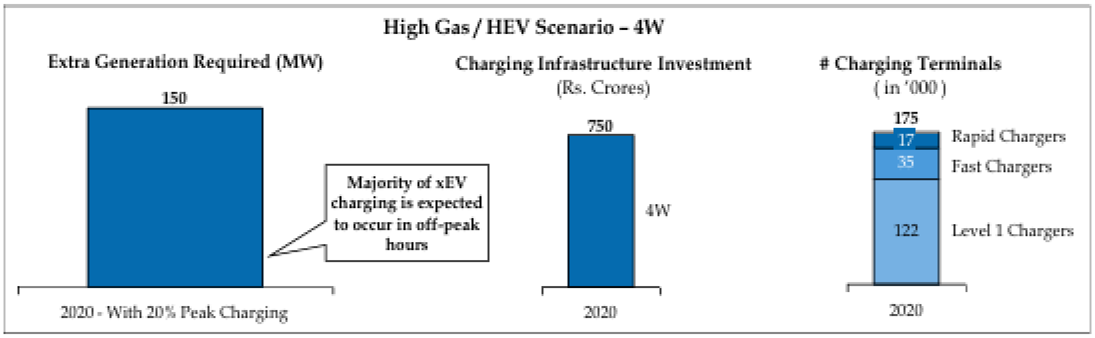
Industry Perspective: Cost-Efficiency and the EV Battery Supply Chain in India

The central determinant of cost-efficiency in India's electric vehicle (EV) industry lies in the battery supply chain. Batteries constitute 35–40% of the overall cost of an electric vehicle, making them the single largest component influencing price competitiveness and adoption rates (Ahmad & Krishna, 2024). For manufacturers, the challenge of cost-efficiency extends beyond simple production costs to encompass raw material sourcing, technological innovation, logistics management, and end-of-life recycling (Ray & Mukherji, 2025a). At present, India remains heavily reliant on imports of critical minerals such as lithium, cobalt, and nickel (Hangzo, 2025), which exposes the industry to volatile global markets and geopolitical risks (Jones et al., 2023). As the sector expands, addressing supply chain vulnerabilities and achieving domestic self-sufficiency in battery production will be crucial for realizing cost-efficiency and long-term sustainability (A. Jha et al., 2022). The global lithium supply chain is highly concentrated, with Australia, Chile, and China controlling the majority of reserves and refining capacity (Ren et al., 2024). Similarly, cobalt production is dominated by the Democratic Republic of Congo, while nickel is largely sourced from Indonesia and the Philippines (Severson et al., 2023). This concentration creates supply risks for Indian manufacturers, who must compete with global giants such as China, the EU, and the U.S (Badri Narayanan G. et al., 2025). A Monte Carlo simulation conducted on the Indian EV battery supply chain estimated that the average supply chain cost stands at USD 505 million, with significant uncertainty arising from raw material shortages and price volatility (Badri Narayanan G. et al., 2025). These costs directly impact vehicle affordability and delay the achievement of economies of scale (Busch et al., 2024). Furthermore, disruptions such as the COVID-19 pandemic and the Russia–Ukraine war have underscored the vulnerability of global supply chains (Allam et al., 2022), adding urgency to India’s pursuit of domestic solutions (Dhairiyasamy et al., 2024).



Sales of Electronic Vehicles across 2014-2023

The Production Linked Incentive (PLI) scheme for Advanced Chemistry Cells (ACC), incentivizes gigafactory development in India, offering financial rewards to firms that achieve high levels of value addition and import substitution (Sarkar et al., 2018). Leading players such as Reliance New Energy Solar, Ola Electric, and Rajesh Exports have committed to setting up multi-GWh battery manufacturing facilities under the PLI scheme (Hemavathi, 2025). These initiatives are expected to reduce per-unit costs by localizing production, generating economies of scale, and building resilience against external shocks (Bose et al., 2024). A promising development in reducing dependency is the discovery of lithium reserves in Jammu and Kashmir in 2023, estimated at 5.9 million tonnes, these reserves could significantly reduce India's import reliance in the medium term, altering the economics of domestic battery manufacturing (Moerenhout et al., 2022). However, mineral extraction requires time, environmental clearances, and technological expertise, meaning short-term reliance on imports will persist (A. Singh, Shankar, et al., 2025). In the meantime, the focus must remain on diversifying sourcing channels and negotiating strategic partnerships with resource-rich countries to secure supplies (Majid et al., 2024).



Infrastructure requirement for xEV four-wheelers

Source: (Ministry of Heavy Industries & Public Enterprises, 2012)

Technological innovation in battery chemistry is another crucial driver of cost-efficiency (Turcheniuk et al., 2021). Presently, lithium-ion batteries dominate the market, but their cost trajectory is shaped by raw material availability and production efficiency (Ipek et al., 2025; Manzetti & Mariasiu, 2015; Meister et al., 2016). Emerging chemistries such as sodium-ion and solid-state batteries hold potential to disrupt this cost structure (Dong et al., 2023; Islam et al., 2024). Sodium-ion batteries, for instance, are less resource-intensive and rely on widely available raw materials such as sodium and iron, making them inherently more cost-efficient and geopolitically secure (Nithya & Gopukumar, 2015; Paul et al., 2024; Phogat et al., 2025; Y. Singh et al., 2022). Solid-state batteries, while currently expensive, promise higher energy density, faster charging, and longer lifespans, which could lower lifecycle costs despite high upfront production expenses (Bobba et al., 2024; A. K. Das et al., 2024). Indian firms, in collaboration with global research institutions, are beginning to invest in these next-generation technologies, which could provide a leapfrogging opportunity for the domestic industry (Retna Kumar & Shrimali, 2020; Thomas et al., 2024). Yet, despite these advancements, the Indian battery supply chain faces persistent challenges that undermine competitiveness compared to China and the EU (Gandhok & Manthri, 2023). Fragmented logistics acts as an impediment because movement of raw materials and finished products is hampered by high transportation costs, port congestion, and inadequate multimodal integration (Ramachandran et al., 2025; Saxena, 2022). While Chinese firms benefit from highly integrated industrial clusters where mining, refining, and cell production are co-located, reducing transaction and transportation costs (X. Wang et al., 2022). Without similar efficiencies, Indian manufacturers face higher overheads that erode cost competitiveness (Sankaran & S, 2022). Another structural weakness is the absence of a robust recycling ecosystem (Kala & Mishra, 2021). Recycling is critical for reducing raw material dependency, lowering costs, and minimizing environmental externalities (Molla et al., 2023). Globally, recycling can recover up to 95% of valuable materials such as lithium, cobalt, and nickel from used batteries (A. Singh & Saxena, 2024). In India, however, recycling remains nascent and fragmented, with only a handful of formal operators and significant dominance of the informal sector (Panda et al., 2023). The lack of standardized collection mechanisms, advanced recycling technologies, and regulatory clarity hinders the ability of the industry to reclaim value from end-of-life batteries (Loganathan et al., 2022). As EV adoption scales, inadequate recycling will not only exacerbate raw material pressures but also create environmental hazards, undermining both cost-efficiency and sustainability (Balaji et al., 2023). Skill shortages further constrain the battery supply chain (Rajendran et al., 2025). The industry requires specialized skills in electrochemistry, materials science, and advanced manufacturing, areas where India faces a significant talent gap (Ray & Mukherji, 2025e). China and South Korea, by contrast, have invested heavily in vocational training and academic-industry partnerships to create a pipeline of skilled labor (Chen et al., 2022). In India, initiatives such as the Skill Council for Green Jobs and collaborations with academic institutions are beginning to address this gap, but progress remains slow (Pavlova & Singh, 2022). Without a skilled workforce, gigafactories and advanced R&D projects will struggle to achieve optimal productivity, raising per-unit costs and limiting competitiveness (Rangaswamy et al., 2023).

Vehicle Segment	**Perception of xEV compared to ICE (Range)**	**Key Barriers for adoption**	**Sensitivity**
Four Wheelers	xEV worse than ICE - **46%**	Range – **46% (4th)** Charging time – **41% (6th)** Charging infra – **31% (10th)**	For PHEVs – Acquisition price > recharge time in sensitivity analysis. **Inference:** Pilots can be conducted for public fast charging terminals to monitor adoption of xEVs before complete roll out.
Buses	xEV worse than ICE - **39%**	Charging time – **56% (3rd)** Charging infra – **45% (7th)** Range – **33% (4th)**	**Recharge time** is the 3rd most important parameters as per sensitivity analysis. **Inference:** Moderate sensitivity observed for recharge time for PHEV and BEV buses indicating the need to set up fast and rapid recharging stations.
Two Wheelers	BEV scooters worse than ICE - **46%** BEV bikes worse than ICE - **46%**	Range – **e-scooters** - **46% (4th)**, **e-bikes – (48%) (4th)** Charging time – **e-scooters - 46% (5th), e-bikes – 43% (5th)** Charging infra – **e-scooters - 30% (10th), e-bikes – 29% (10th)**	**Recharge time** and **range** are the 2nd and 4th most important parameters as per the sensitivity analysis. **Inference:** Typical changes to re-change time translates to significant variations in consumer preference for BEV bikes and scooters, while the consumers are relatively agnostic to changes in range of the vehicle.
Three Wheelers	xEV worse than ICE - **46%**	Range – **50% (4th)** Charging time – **50% (4th)** Charging infra – **42% (8th)**	**Recharge time** is the 3rd most important parameters as per sensitivity analysis.

Key xEV charging infrastructure related findings from the consumer survey

Source: (Ministry of Heavy Industries & Public Enterprises, 2012)

From an industry perspective, the path to cost-efficiency also requires integration with renewable energy systems (Sinha et al., 2023). The carbon footprint of battery production is often criticized, and high electricity costs in India further inflate production expenses (Bohre et al., 2022; D. Das et al., 2024; P. Jain et al., 2025; Prianka et al., 2024; Sree Lakshmi et al., 2020). By co-locating gigafactories with renewable energy clusters, such as solar and wind parks, manufacturers can reduce both costs and emissions, aligning with global ESG requirements (Banerji et al., 2021). This integration not only enhances cost-efficiency but also strengthens the industry's access to green financing, a growing determinant of competitiveness in international markets (Sawhney & Gambhir, 2024; Wasan et al., 2024). Comparing India with global leaders underscores the magnitude of the challenge. China accounts for over 70% of global lithium-ion battery production capacity, benefiting from state-led industrial policy, large-scale investment, and integrated supply chains (Sun et al., 2021). The European Union has adopted a "battery passport" framework to ensure transparency, sustainability, and circularity in its supply chain (Butera & Gatteschi, 2024). In contrast, India's industry remains at a nascent stage, fragmented across production, assembly, and recycling. To bridge this gap, Indian manufacturers must not only scale up but also adopt international best practices in supply chain transparency, traceability, and ESG compliance (Reynolds, 2024). Despite these challenges, opportunities abound. The Indian EV market is projected to reach USD 7 billion by 2025, creating a massive demand base for domestically produced batteries (Kashyap & Pachwarya, 2022). By leveraging PLI incentives, mineral discoveries, and technological innovation, India has the potential to establish itself as a major hub in the global EV value chain (D. S. Patil et al., 2023). The government's emphasis on creating localized clusters for EV production, akin to special economic zones, can further reduce transaction costs and enhance cost-efficiency (S. Biswas &

Deb, 2024). Moreover, growing investor interest, evidenced by multi-billion-dollar commitments from global and domestic firms, signals strong confidence in the sector's long-term viability (Ray & Mukherji, 2025b).

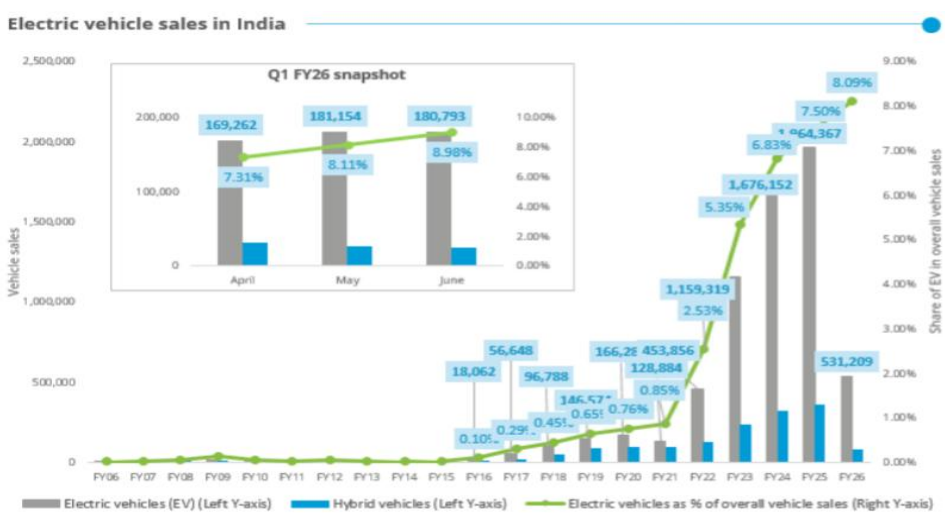
Civil Society Perspective

From the standpoint of civil society, the cost-efficiency of electric vehicles (EVs) in India extends far beyond the simple act of replacing petrol or diesel cars with battery-powered alternatives. It reflects a larger ecosystem of household economics, social health, environmental protection, and community well-being . While Indian consumers often hesitate to adopt EVs due to the higher upfront costs, long-term analysis shows that EVs deliver significant savings through lower operating expenses, reduced maintenance costs, and the availability of fiscal incentives (Weldon et al., 2018). More importantly, at the societal level, EV adoption translates into improved air quality, reduced dependence on fossil fuels, job creation, and enhanced social equity. (Rajagopal, 2023) These benefits are distributed across multiple layers of Indian society, making the EV transition a critical pillar of sustainable development and an essential tool for reshaping the mobility landscape of the country (Mittal et al., 2024). At the household level, the cost advantage of EVs becomes clearer when lifetime ownership is considered (Huber et al., 2021). Although EVs carry a higher purchase price because of battery expenses, their operating and maintenance costs are substantially lower, studies suggest 60–70% less compared to internal combustion engine (ICE) vehicles. Electricity in India costs around ₹6–8 per kWh, which translates to a running cost of about ₹1–1.5 per kilometer, whereas petrol vehicles cost about ₹5.5 per kilometer, hence, for a typical driver covering 1,000 kilometers per month, this amounts to annual savings of nearly ₹48,000 on fuel alone. Over a five-year period, savings from fuel and maintenance combined can exceed ₹1.5 lakh, even when factoring in battery depreciation and replacement costs. Government subsidies narrow the affordability gap further. Under the FAME-II scheme, buyers can receive incentives up to ₹1.5 lakh, and state governments such as Delhi and Maharashtra provide additional benefits, including exemptions on road tax and registration fees, or subsidies worth up to ₹2.5 lakh for certain models (Kohli, 2024). These policies together ensure that EVs become financially viable options for India's middle-class households, thereby reinforcing their long-term cost-efficiency (Ghosh & Sarkar, 2022).

Subsidies and exemptions impact on the final price of an EV

Source: (Nair & Minocha, 2023)

The broader health and environmental benefits of EV adoption further enhance their value to society. India ranks among the most polluted nations globally, with 14 of the world's 20 most polluted cities located within its borders, and vehicular emissions constitute a major share of this burden. Tailpipe emissions from ICE vehicles release nitrogen oxides, volatile organic compounds, and fine particulate matter (PM2.5), all of which contribute to chronic respiratory and cardiovascular illnesses . EVs, by contrast, produce zero tailpipe emissions, and widespread adoption could potentially reduce carbon dioxide emissions by more than 200 million tonnes annually by 2030. Such a reduction directly translates into fewer cases of asthma, lung cancer, and other pollution-related diseases, which in turn lowers both household medical expenditures and the government's healthcare subsidies. Studies have estimated that air pollution costs India nearly 1.2% of its GDP annually through increased healthcare expenses and productivity losses, underscoring the scale of potential savings. Furthermore, EVs reduce noise pollution, which, although less visible than air quality issues, has significant effects on mental health, quality of life, and stress levels in urban communities. The quieter operation of EVs fosters calmer residential environments and improves urban livability, particularly in densely populated areas. Employment generation adds another layer to the civil society benefits of EV adoption. India's EV sector is projected to create up to 10 million direct jobs and 50 million indirect jobs by 2030. These opportunities will span across manufacturing, assembly, software development, charging infrastructure, battery production, and recycling industries. Such a broad labor market transformation will not only provide new sources of livelihood for millions of Indians but will also support industrial diversification, which is critical for economic resilience. Many of these jobs will arise outside metropolitan hubs, as gigafactories, renewable energy parks, and charging networks spread into semi-urban and rural regions. This decentralization supports more equitable development across states. Skill development initiatives by the Skill Council for Green Jobs and partnerships between government and educational institutions are already preparing a new workforce adept in electrochemistry, battery maintenance, and energy management (Rangaswamy et al., 2023). For civil society, these job creation opportunities represent a crucial benefit, as they directly strengthen household income security while also enhancing India's standing in global value chains. The affordability of EVs also carries implications for social equity. Rising petrol prices disproportionately affect low- and middle-income households, since transportation costs account for a larger share of their disposable income. By reducing dependence on expensive and volatile fossil fuels, EVs cushion vulnerable groups from global oil price shocks. Auto-rickshaw drivers and two-wheeler commuters in particular stand to benefit, since their daily livelihoods depend heavily on mobility costs (Mishra et al., 2024). Already, electric scooters and e-rickshaws dominate last-mile connectivity in several Indian states, delivering cost savings to drivers while offering affordable fares to passengers (Sasidharan & Das, 2022). EVs may also indirectly improve gender equity, as studies suggest women are more inclined to adopt vehicles that are perceived as low maintenance and safe. Thus, EV adoption can expand mobility opportunities for groups historically marginalized in India's transport systems, thereby contributing to a more equitable and inclusive society.



Electric Vehicles Sales in India

Source: (Nair & Minocha, 2023)

EVs offer smoother acceleration, instant torque, and quieter operation compared to ICE vehicles. Many models include advanced features such as regenerative braking, AI-assisted navigation, and smartphone integration, which enhance user convenience. For tech-savvy youth, this appeal of advanced features goes beyond financial savings and creates a cultural enthusiasm for cleaner technology. The popularity of EVs among early adopters also generates social reinforcement, as owning an EV is increasingly associated with environmental responsibility and higher social standing. This “green prestige” strengthens peer-driven diffusion of EVs within communities and accelerates societal acceptance. Beyond personal ownership, EV adoption has the potential to transform community life and urban planning. As residential complexes, shopping centers, and workplaces begin installing charging infrastructure, mobility becomes integrated with decentralized energy systems. Solar-powered charging hubs, for example, allow households to generate and consume their own clean electricity, creating energy independence at the community level. This integration between renewable energy and EVs empowers communities to take charge of their own energy futures and reduces reliance on centralized, fossil-fuel-dependent power grids. Civil society thus benefits not only through lower costs but also through greater resilience against energy shocks and environmental crises. A critical area of societal benefit relates to recycling and the circular economy. While India’s battery recycling industry remains nascent, it has the potential to recover up to 95% of valuable materials such as lithium, cobalt, and nickel from used batteries. Building such an ecosystem would reduce raw material dependency, lower production costs, and create localized jobs in recycling industries. For civil society, this means safer waste management practices, reduced environmental hazards, and improved livelihoods for informal sector workers who can be transitioned into formal, regulated recycling industries. Recycling also prevents hazardous waste accumulation, ensuring environmental justice for vulnerable populations often exposed to industrial pollution.

Conclusion

A careful balancing act between short-term financial costs and long-term national benefits is reflected in the government's position on EV adoption. Significant upfront investments are made in the form of FAME I and II subsidies, PLI scheme industrial incentives, and tax rationalization initiatives like GST reductions. The savings from fewer oil imports, lower medical costs, improved energy security, and higher industrial competitiveness, however, more than offset these costs. Because the EV transition converts spending into strategic investments with compound economic and social returns, rather than cutting government spending in the short term, it is therefore cost-effective. Designing policies that progressively lessen reliance on subsidies while maintaining fair access, robust localization, and resilience in vital supply chains will be India's future challenge. Designing policies that progressively lessen reliance on subsidies while maintaining fair access, robust localization, and resilience in vital supply chains will be a challenge for India. If successfully implemented, the government's EV initiatives will establish India as a global leader in sustainable mobility while also proving to be financially responsible. The government's financial commitments to EV adoption should be viewed as strategic investments that yield multifaceted returns rather than as expenses. The government benefits from increased tax revenues, lower oil import costs, job creation, healthcare savings, industrial growth, and improved geopolitical standing once the EV market matures. These benefits greatly exceed the original tax breaks and subsidies, demonstrating that India's EV policy is not only morally and environmentally sound but also wise financially in the long run. India can obtain its position as a global leader in sustainable mobility and optimize its return on investment from the EV transition by guaranteeing effective implementation, fair distribution, and strategic localization. The battery supply chain is critical to the EV industry's cost-effectiveness. Reliance on imported minerals, global market volatility, logistical inefficiencies, and batteries which can account for as much as 40% of vehicle costs present vital obstacles. However, there are ways to become more self-reliant thanks to government initiatives like PLI incentives, the discovery of domestic reserves, and gigafactory investments. If successfully marketed, emerging chemistries like solid-state and sodium-ion batteries could further improve cost-efficiency. To be competitive with world leaders like China and the EU, however, enduring problems like poor recycling infrastructure, disjointed logistics, and a lack of skilled workers must be resolved. Whether the Indian EV industry can become a cost-effective, globally competitive player in the electrification era ultimately depends on its capacity to strike a balance between short-term expenses and long-term benefits. Adoption of EVs is a sign of a larger social shift rather than just a change in technology. It changes perceptions of consumption, energy use, and mobility while fostering a new social consciousness around sustainability. Lower operating expenses benefit households, better air quality benefits communities, and new job opportunities benefit workers. Even though the adoption costs may seem high at first, these cumulative benefits create a very positive cost-benefit ratio for society. Affordability will continue to improve, speeding up adoption rates, as battery prices have dropped by almost 89% globally between 2010 and 2022 and are predicted to continue declining. Consequently, the long-term cost-efficiency of EVs is clearly favorable from the perspective of civil society.

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